SENTINELWIFI / PASSIVE WIRELESS SECURITY RESEARCH

Machine Learning-Based Wireless Intrusion Detection System

Detection of Deauthentication Flood Attacks Using Random Forest

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01 Introduction & problem

Wi-Fi management messages control connection state. Concentrated deauthentication traffic can disrupt availability. Passive monitoring can expose abnormal frame rates without decrypting user payloads or transmitting attack traffic.

02 Objectives

Identify IEEE 802.11 frames; extract meaningful rolling features; classify normal and malicious observations; detect floods; persist explainable alerts; evaluate on held-out labels.

03 Methodology

Data: The WPA3 Dataset, Deauth subset (MIT). 122,306 publisher-labeled packets: 116,799 Normal and 5,507 Deauth. AWID3 access has been requested.

Processing: Stream CSV chunks through the same metadata and rolling-feature representation used for PCAP inference. Group by observed transmitter/BSSID; never train on MAC identifiers.

Evaluation: Purged 60-second blocks; fixed seed-42 grouped holdout; train-only imputation. 3,484 training and 2,802 test windows. One recording, not independent sessions.

04 Detection features

18 extracted measurements; **16 verified classifier inputs**: frame/deauth rates, subtypes, retry ratios, destination diversity, mean size, signal coverage and reason codes. RSSI mean/variability remain analysis-only because antenna selection differs.

Default rule: **10 deauth/s over 5 seconds**; 30-second cooldown. Tune for the environment.

05 System architecture

Wireless PCAP / authorized live capture
Scapy: raw 802.11 metadata

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Rolling windows & canonical features
5-second window / 1-second step

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Complementary detection engine
Rate rule + compatible Random Forest

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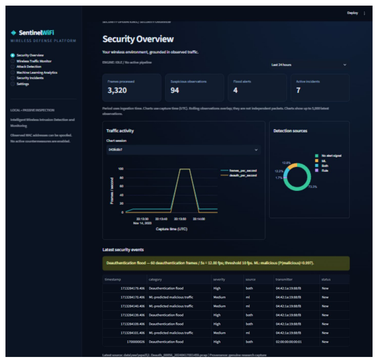
SQLite alerts & observations
Evidence, source, severity and status

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Streamlit dashboard & exports
Monitor, investigate and report

Separate training path: Labeled packet CSV → same features → purged train/test split → evaluation → saved pipeline. Training labels never enter inference.

06 Working implementation



Actual Streamlit application with persisted packet-derived evidence. Replay is synthetic software-test traffic; measured model results use genuine publisher-labeled packets.

07 Conclusion

The prototype connects real packet parsing, rolling detection, automatic SQLite incidents and an operational dashboard. Offline replay permits repeatable demonstrations without radio hardware. No WIPS enforcement is claimed.

08 Measured ML results

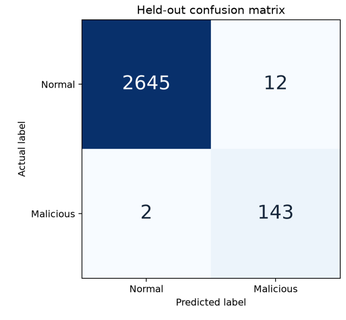
99.50%
Accuracy

92.26%
Precision

98.62%
Recall

95.33%
F1-score

Held-out confusion matrix
Rows: actual; columns: predicted. Normal then Malicious.



12 false positives; 2 false negatives in this holdout. False positives consume investigation time; missed attacks hide threats. These results do not establish performance on other networks.

09 Limitations & future work

Evaluation: Single recording; overlapping within-block observations. 1,023 test feature rows repeat training values, reducing independence.

Wireless: MAC spoofing, packet loss, protected frame visibility and radio noise. Linux live capture needs monitor-mode hardware; not tested on this Windows host.

Next: Independent captures, approved AWID3 comparison, false-positive tuning and approved AP integration. No packet injection.

10 References & disclosure

[1] HSNG (2024), The WPA3 Dataset. github.com/hsngccu/WPA3-dataset, MIT. Revision e07113327e20.
[2] Chatzoglou et al. (2021), AWID3. IEEE Access 9. DOI: 10.1109/ACCESS.2021.3061609.
[3] Scapy Dot11 and Scikit-learn documentation.

OpenAI Codex assisted implementation, tests and documentation. Student review and demonstration remain required.